Global Reporting Regulation, Stock Liquidity, and Policy Diffusion

Abstract

We study stock liquidity around 17 families of reporting regulation and their international diffusion. We distinguish national implementation from issuer eligibility and actual public reporting. The market panel contains 1,055,142 security-years in 90 jurisdictions. Source-supported comparisons cover seven first-year families and five balanced-window families. Their first-year log-spread contrasts are -0.077 with equal family weights and +0.187 with market-capitalisation family weights. Audit reporting receives 85.6% of the latter weight, equivalent to only 1.36 equally weighted families in concentration terms. An exact decomposition locates the difference in policy composition. First statutory reporting also follows voluntary disclosure in documented cases, limiting the information increment. Complementary liquidity outcomes and pre-policy differences constrain a general causal benefit. No diffusion coefficient survives broad multiplicity adjustment. The evidence identifies the importance of the policy mix, covered population and information timing when comparing reporting mandates.

1. Introduction

Reporting regulation can reduce adverse selection by supplying credible information. Its market effect should depend on the incremental information, the firms required to disclose and the audience receiving it. We ask how stock liquidity changes around reporting mandates and whether foreign policy choices help explain their diffusion. Combining mandates requires a further choice: whether the object is the average regulation family or the regulation exposure of large issuers.

Following the empirical logic of Krueger et al. (2023), we establish obligations and reporting dates before connecting them to issuer outcomes. The setting extends beyond comprehensive sustainability reporting to 17 economic regulation families. We then apply the competition, learning, emulation and coercion framework of Koning, Mertens and Roosenboom (2018). A national implementation date establishes geographic exposure; a covered-issuer study additionally needs evidence of the relevant issuer obligation or report.

The principal finding is that policy composition materially changes the aggregate spread contrast. Across seven source-supported families, equal-family and market-capitalisation weights yield -0.077 and +0.187, respectively. Audit reporting accounts for 85.6% of market-cap weight. We decompose the difference exactly and disclose every single-family exclusion, rather than select a favourable omission. First-year and longer-window support differ. The estimates therefore describe specified policy mixes, not a common causal reporting benefit.

2. Institutional setting and research design

A public report can change investor information; confidential CbCR submitted to tax authorities does not constitute the same public-information event. Comprehensive mandatory E/S/G requires a response on all three topics, with the final topic determining the joint date. CSR spending, climate-only rules and voluntary guidance are classified separately. The inherited topic review retains 37 qualifying phases from 41 country reviews; earliest separate topic histories remain incomplete. Bank public CbCR attaches to institutions and consolidation perimeters, not automatically every listed holding company.

Eligibility and information timing are rule specific (Table 1). The source-supported studies cover auditor reports, XBRL, conflict minerals, bank CbCR, IFRS, German statutory nonfinancial reports and extractive payments. BP, Total SA and Tullow report FY 2015 statutory payments in 2016. BP publishes on 24 June. Total states that its French registration document was filed on 16 March; the signature is dated 15 March. A contemporaneous SEC rule corroborates Tullow's 15 March annual report. Tullow explicitly reports voluntarily since 2013. Rio Tinto's June voluntary Taxes paid release does not establish the separate statutory filing date, which remains unresolved.

The 2000-2025 panel contains 99,100 vendor quote/company identifiers. Outcomes are log median relative quoted spread, log median Amihud price impact and the fraction of valid observed returns equal to zero. Each requires 120 valid observations; logged proxies use annual 1/99 winsorisation. These calendar-year measures differ from the reference study. Exact historical home-share identifiers and source reports corroborate the selected issuer matches; current company names alone do not establish historical legal entities.

For each family, cohort and endpoint, we subtract the weighted comparison change from the implementer change relative to g -1. Geographic comparisons require known later implementation, at least 30 issuers per country and three comparison markets. Source-supported comparisons require at least three treated and 20 comparison issuers. First-year contrasts use g -1 and g. Balanced support is fixed from g -3 through g +2 and averages years 0-2. Unknown histories are not coded never-treated. Few legal waves, selection and cross-market differences limit policy inference.

GDP in g -1 is allocated over the fixed source-study issuer universe within each country; cohorts are weighted by treated mass. Across economic families, equal weights assign 1/F and market-cap weights use pre-event treated issuer capitalisation. Both use the same supported family set for all outcomes, and banking enters once. Equal-issuer and issuer-cap within-family alternatives are disclosed. Market-cap exposure counts repeated issuer-policy observations and is not a unique-issuer portfolio.

Source candidates are selected by pre-event size before new coefficients are inspected. IFRS comparisons require US domestic-bank and auditor US GAAP evidence. German nonfinancial-report comparisons require US parent incorporation. Extractive comparisons use resource-sector parents with contemporaneous US incorporation/listing evidence, withholding direct foreign-regime flags. US payment rules were not operative in the 2016-18 window, but foreign subsidiaries and voluntary reporting remain possible. We review 164 additional full SEC filings; source and outcome failures remain visible. Issuer sampling intervals preserve shared-issuer covariance and do not supply causal policy confidence intervals.

3. Liquidity, information timing and policy composition

Table 2 Panel A reports geographic exposure. Fifteen first-year families yield equal-family spread -0.049 and price impact -0.072, compared with +0.015 and +0.009 under market-cap weights. Confidential tax CbCR receives 33.1% of that exposure weight. Nine families support the balanced window. Significant pre-policy spread differences and unresolved eligibility prevent interpreting these broad contrasts as effects on all legally covered issuers.

Panel B uses seven source-supported first-year families. Equal-family spread is -0.077; market-cap spread is +0.187. Price impact is +0.025 and +0.190, respectively. Figure 1 reports every family and its market-cap weight. The cap-weighted sample is concentrated: its effective family count, 1/sum(w squared), is 1.36. The cap-minus-equal spread difference is +0.264 log points. The downloadable decomposition attributes this difference to each family's weight change times its fixed estimate, and reports every leave-one-family-out contrast. Reweighting changes the target population; it does not improve identification.

Table 3 identifies the individual studies. CAM has 1,481 treated and 2,767 comparison issuer-cohort observations, while early XBRL has six treated and 62 comparisons. The IFRS study contains four Canadian banks and 42 US comparisons; German statutory nonfinancial reporting contains three issuers and 35 comparisons. Their first-year spread contrasts are -0.235 and -0.217, but IFRS price impact increases. The source and comparison populations differ across these settings.

Extractive reporting extends support beyond six families. BP, Total and Tullow have three treated and 35 first-year comparisons, with spread +0.004, price impact +0.162 and zero-return fraction +0.00334. Balanced support retains 24 comparisons. Prior voluntary reporting at Tullow means the statutory cycle need not create wholly new information. Commodity prices, Brexit, the BP settlement and cross-country exposure also change in this window. We retain this study under both aggregate weights because source support, rather than the coefficient sign, determines inclusion.

Five families support balanced years 0-2: banks, conflict minerals, IFRS, German nonfinancial and extractive reports. Equal-family and market-cap spreads are -0.042 and -0.058, while price impact is +0.149 and +0.211. Figure 2 shows the fixed-sample profiles and material pre-policy differences. The shorter-window seven-family aggregate cannot be joined to this five-family series as a single dynamic effect. The evidence supports studying information increments and policy composition, while leaving a universal causal liquidity improvement unestablished.

4. International diffusion and conclusion

Following Koning et al., competition uses prior adoption among export-pattern peers; learning uses prior outcomes among geographic neighbours; emulation uses language, legal-origin and IGO peers; external dependence uses IMF credit and aid. Signals precede the focal decision. Family-year strata and country-clustered covariance accompany timing and status alternatives, with lagged income, growth and trade controls where specified. The ledger records documented phases rather than certified first-ever adoption.

Across 1,330 attempted designs, 433 stable focal coefficients remain and none survives the broad Holm correction. We refit the full matrix; additional issuer report dates do not create new national adoption decisions. Later liquidity estimates are not inserted into earlier learning signals. Reporting mandates differ in their information increments and covered economic mass. A comparison of those differences provides a coherent research question, while source coverage, pretrends and common legal waves constrain causal and diffusion claims.

Table 1. Implemented regulation families and issuer evidence

Economic regulation family	Geographic support First / balanced	Covered-issuer evidence / remaining limit
Audit oversight / rotation	Yes / Yes	Auditor/client history needed
Expanded auditor reporting	Yes / Yes	US CAM filing cohorts
Beneficial ownership	Yes / Yes	Listed exemptions; venue gaps
Banking public CbCR	Yes / Yes	Bank perimeter; 3 dated publishers
Confidential tax CbCR	Yes / Yes	Nonpublic returns; UPE gaps
General public tax CbCR	Yes / No	No observed eligible filing cohort
Climate disclosure	Yes / Yes	Historical thresholds/report gaps
Sectoral climate reporting	No / No	Sector and panel boundary
Digital assets	Yes / Yes	Crypto object; equity scope differs
Comprehensive mandatory E/S/G	Yes / Yes	3 German statutory report publishers
Extractive payments	Yes / No	3 statutory reporters; foreign-scope gaps
Corporate governance	Yes / No	Listing and size phases missing
IFRS reporting	Yes / No	4 dated Canadian first IFRS reporters
Supply-chain reporting	Yes / Yes	3 US first-cycle reporters
Financial sustainability disclosure	Yes / No	Late phases; actual reports missing
Informed sustainability reporting	No / No	Scope and calendar unsupported
XBRL / digital filing	Yes / No	US early phase; first exhibit unknown

Coverage through 2025 only; future schedules are excluded. Geographic support is outcome support, not verified legal eligibility. Covered issuer studies support seven families in the first year and five in the balanced observed-report subset. Country and issuer thresholds, exemptions, reporting periods and actual publication dates remain separate. Full legal scopes and event-level gap overlays are online.

Table 2. Both cross-family weighting schemes

Population / coverage / outcome	Equal family Estimate [interval]	Market-cap family Estimate [interval]
A. Geographic issuer exposure First year (15 families): Log spread	-0.049 [-0.082, -0.016]	+0.015 [-0.027, +0.056]
A. Geographic issuer exposure First year (15 families): Log price impact	-0.072 [-0.115, -0.030]	+0.009 [-0.059, +0.078]
A. Geographic issuer exposure First year (15 families): Zero-return fraction	-0.0076 [-0.0105, -0.0048]	-0.0025 [-0.0063, +0.0014]
A. Geographic issuer exposure Years 0-2 (9 families): Log spread	-0.032 [-0.088, +0.023]	+0.040 [-0.039, +0.120]
A. Geographic issuer exposure Years 0-2 (9 families): Log price impact	-0.013 [-0.080, +0.054]	+0.095 [-0.025, +0.215]
A. Geographic issuer exposure Years 0-2 (9 families): Zero-return fraction	-0.0018 [-0.0077, +0.0040]	+0.0067 [-0.0073, +0.0206]
B. Source-corroborated subset First year (7 families): Log spread	-0.077 [-0.125, -0.028]	+0.187 [+0.150, +0.224]
B. Source-corroborated subset First year (7 families): Log price impact	+0.025 [-0.074, +0.124]	+0.190 [+0.128, +0.252]
B. Source-corroborated subset First year (7 families): Zero-return fraction	+0.0014 [-0.0031, +0.0060]	+0.0057 [+0.0036, +0.0079]
B. Source-corroborated subset Years 0-2 (5 families): Log spread	-0.042 [-0.117, +0.033]	-0.058 [-0.126, +0.010]
B. Source-corroborated subset Years 0-2 (5 families): Log price impact	+0.149 [+0.040, +0.258]	+0.211 [+0.115, +0.307]
B. Source-corroborated subset Years 0-2 (5 families): Zero-return fraction	+0.0005 [-0.0051, +0.0061]	+0.0016 [-0.0030, +0.0062]

GDP weights within families. Panel A intervals condition on country sampling and the observed policy calendar; Panel B intervals describe issuer sampling only and are not causal policy confidence intervals. Zero returns are fractions. Samples are fixed within each horizon; family composition differs across horizons and populations. Equal-issuer/cap within-family alternatives and full diagnostics are online. No unsupported family is assigned a zero estimate.

Table 3. Covered-issuer first-year comparisons beyond banking

Source-corroborated study	Treated / comparison issuer-cohort observations	Log spread	Log price impact	Zero-return fraction
US auditor CAM (2019-21)	1,481 / 2,767	+0.236	+0.207	+0.0063
US early XBRL (2009)	6 / 62	-0.329	-0.417	-0.0004
US conflict minerals (2014)	3 / 70	-0.085	+0.249	+0.0071
Observed bank full reports (2015)	3 / 21	+0.089	+0.077	+0.0155
Canadian first IFRS reports (2012)	4 / 42	-0.235	+0.132	-0.0096
German statutory E/S/G (2018)	3 / 35	-0.217	-0.235	-0.0124
UK/FR statutory extractive reports (2016)	3 / 35	+0.004	+0.162	+0.0033
Alternative: bank initial phase (2014)	23 / 21	+0.035	-0.343	+0.0006

Counts are issuer-cohort observations; comparison issuers can recur across CAM cohorts. GDP is allocated over the fixed study country universe. The observed-report aggregate uses the first seven rows. The last row replaces banking full reports in a separate sensitivity. Sparse, selected treated groups, industry differences, common waves and subsidiary spillovers limit interpretation; REIT comparisons are industry screened rather than legally certified noncovered. IFRS controls require current domestic-parent/US-GAAP evidence; German E/S/G controls can have covered subsidiaries. Full estimates retain balanced and alternative weights.

Source review and timing

The all-data audit indexes 12,326 PDF entries. Three additional local issuer reports and archived statutory reports support the extractive review; original documents, mirrors and blocked primary downloads are separately recorded. The control review adds 164 full SEC filings to 402 prior filings. BP, Total and Tullow dates are corroborated; Rio's statutory date remains pending. Readable selected reports did not require OCR. Historical issuer thresholds, subsidiary perimeters and first-report histories remain incomplete.

References and primary sources

Krueger, P., Z. Sautner, D. Y. Tang and R. Zhong. 2023. The Effects of Mandatory ESG Disclosure Around the World. Supplied working paper; Journal of Accounting Research (2024), doi:10.1111/1475-679X.12548.

Koning, M., G. Mertens and P. Roosenboom. 2018. Drivers of institutional change around the world: The case of IFRS. Journal of International Business Studies. Supplied article.

BP, Total SA and Tullow Oil. FY2015 statutory payments disclosures. BP SEC 6-K accession 0001191638-16-002266, exhibit 1.4; Total 2015 Registration Document, filing certification and statutory payment chapter; Tullow 2015 Annual Report, pp 171-175. Exact URLs and hashes accompany the online issuer evidence.

SEC. Disclosure of Payments by Resource Extraction Issuers. Releases 34-78167 (2016), footnote 302, and 34-90679 (2020), prior-rule history. UK SI 2014/3209; French Commercial Code L225-102-3. Source reviews distinguish statutory, voluntary and confidential tax reporting.

Canadian banks: first-quarter 2012 issuer releases and Bank of Canada/OSFI transition notes. German statutory reporting: BASF, Bayer and BMW 2017 reports, public 2018. Original sources and exact publication precision are retained in the companion register.

PCAOB AS 3101; SEC Interactive Data CDI Q105.04; SEC Conflict Minerals Release 34-67716; inherited bank institution/perimeter review. Original source records, eligibility failures and alternative estimates remain available in the companion evidence.

Figure 1. Seven source-supported regulation families

First-year GDP-weighted issuer comparisons. Bars describe conditional issuer sampling, not causal policy confidence intervals. Every supported family appears; banking enters once. Right panel shows pre-event market-cap family weights. Zero-return outcomes are fractions. Small treated groups and different industry/country exposures remain material.

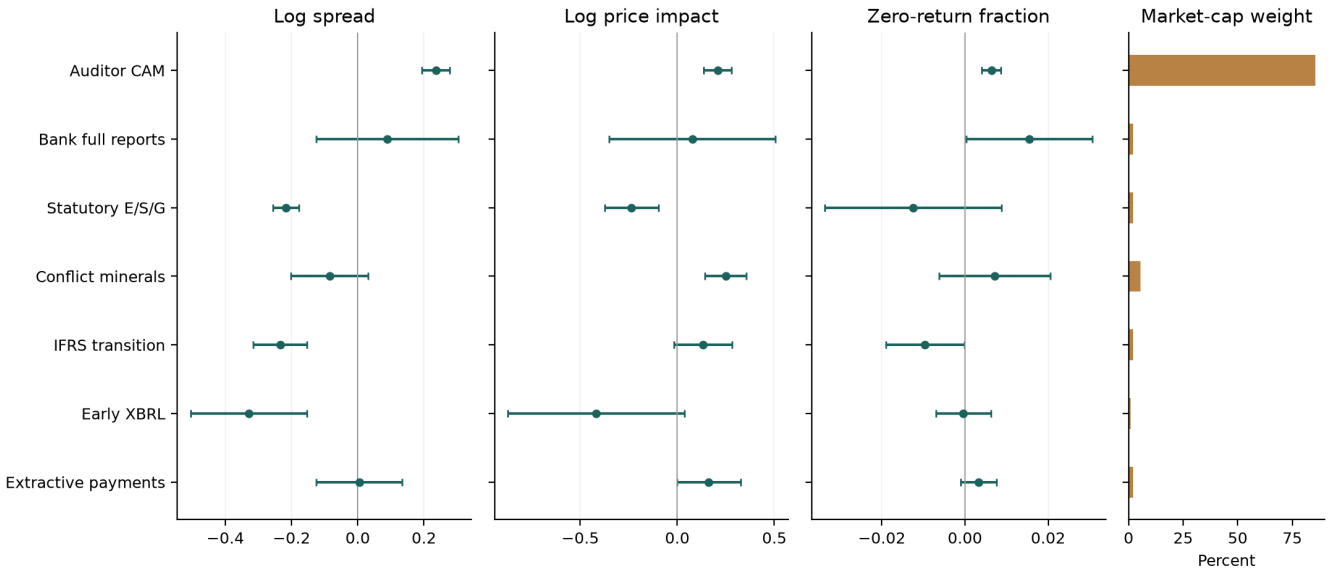
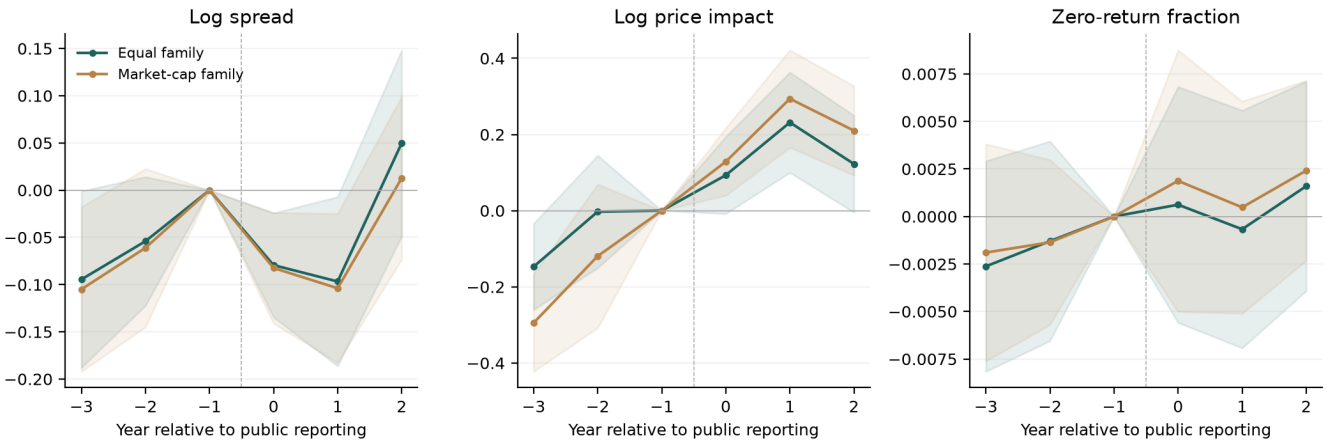


Figure 2. Five-family source-supported balanced profiles

Fixed g-3 through g+2 support: observed bank reports, conflict minerals, Canadian IFRS, German statutory nonfinancial and extractive reports. Year -1 is zero. Shading describes pointwise issuer sampling, with shared-issuer covariance retained across families. These sparse collective policy waves do not support causal policy inference. The broad geographic panel has separate adverse lead diagnostics.



Reading the exhibits

The broad exposure panel and the source-corroborated studies answer different questions. The first tracks issuer outcomes around geographic regulation calendars; the second conditions on specific legal or report evidence. The seven-family first-year and five-family balanced estimates differ in both composition and horizon. A favourable geographic coefficient cannot close an eligibility gap. First-year and balanced windows also change support, and should not be joined into one event path.

Reproducible evidence

The companion report provides the frozen design, all family and aggregate estimates, weight contributions, coverage failures, alternative weights, source dispositions and 441 additive event-review JSONs. Licensed issuer and daily market data remain local. The complete diffusion matrix includes failures and coefficients opposing the hypothesised mechanisms. https://hunternbh.github.io/policy-diffusion/